

Does crop-host supervision improve Swin-based multi-crop condition recognition?

A controlled 90-run in-domain audit of performance, consistency, and confidence

1 Identity-controlled data

NPD-Rice-COCO-43 43 classes

MultiCrop-88 88 classes

Agri-Foundation-145k 215 classes

Known-family + exact-identity controls
Locked train / validation / test identities

2 Paired Swin-S training

Flat

condition head

Dual

condition + host

10 paired seeds per primary model

5 matched seeds: MAE and $KL(q||p^h)$

30 epochs per run | 90 runs total

3 Locked-test audit

Accuracy

Macro-F1

CHER

Paired equivalence + HDR + TCI

Family balance | long tail | host-level error

Oracle host | calibration | selective risk

Predicted host supervision

Equivalent on all 43-class primary outcomes

TCI changed mean accuracy by at most 0.018 percentage points

Ground-truth-host oracle

+2.2 points accuracy on the 215-class benchmark

+10.2 to +10.6 points macro-F1, without refitting

Consistency is not correctness

Use hard host constraints only when the host information is independently reliable.